

OWOSSO, MI, USA

WMO: 722346

Lat: 42.993N

Lon: 84.139W

Elev: 736

StdP: 14.31

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 54824

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.1	5.3	-7.7	3.7	1.1	-2.1	5.1	7.1	25.3	22.4	21.9	26.4	4.4	250	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	89.9	71.7	87.1	70.9	84.1	69.3	75.4	83.8	73.7	81.8	72.2	80.1	9.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.0	125.8	79.4	71.6	119.9	78.0	69.9	113.1	76.4	39.4	83.8	37.7	81.4	36.3	80.0	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.6	18.2	16.3	DB	-6.0	93.8	5.9	2.5	-10.3	95.6	-13.8	97.1	-17.1	98.5	-21.4	100.3	(4)
			WB	-6.6	78.2	5.8	1.8	-10.8	79.5	-14.2	80.6	-17.5	81.6	-21.7	82.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.8	24.4	25.1	35.0	46.9	58.8	67.8	71.6	69.6	63.3	51.7	39.7	30.5
	DBStd	18.97	10.85	10.99	11.21	8.95	8.89	6.76	5.85	5.45	7.77	8.83	9.13	8.80
	HDD50	3178	795	698	484	163	17	0	0	2	85	329	605	
	HDD65	6539	1259	1117	930	546	239	43	12	19	125	423	758	1068
	CDD50	2756	1	1	21	69	290	535	670	608	400	139	21	1
	CDD65	642	0	0	1	2	47	128	217	162	73	12	0	0
	CDH74	5902	0	0	14	44	496	1241	2055	1347	602	102	1	0
	CDH80	1746	0	0	2	3	131	385	688	350	169	18	0	0
Wind	WSAvg	6.6	7.9	7.8	7.5	8.0	6.7	5.8	4.9	4.5	4.8	6.4	7.4	7.6
Precipitation	PrecAvg	29.6	1.7	1.3	2.0	2.8	3.0	3.2	2.7	3.2	3.3	2.2	2.6	2.0
	PrecMax	43.0	4.3	3.8	4.5	5.7	10.2	6.1	6.2	9.9	10.3	6.0	6.5	4.4
	PrecMin	20.5	0.2	0.0	0.4	0.6	0.1	0.5	0.4	0.6	0.0	0.4	0.5	0.3
	PrecStd	5.0	1.0	0.9	1.0	1.3	1.9	1.5	1.3	1.7	1.9	1.2	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	58.0	75.1	79.5	88.2	91.3	93.2	90.6	90.4	82.5	69.9	59.4
		MCWB	52.6	51.3	62.5	59.5	69.4	72.2	73.7	72.0	70.0	68.7	56.4	55.8
	2%	DB	48.9	50.2	64.3	73.1	83.6	88.3	90.0	87.1	85.6	76.9	62.8	51.6
		MCWB	45.4	46.1	54.6	56.0	66.6	70.9	72.7	70.9	70.7	63.7	53.1	48.2
	5%	DB	43.1	43.5	57.4	68.7	80.3	84.5	87.0	84.0	80.9	71.9	58.6	46.4
		MCWB	39.6	39.1	50.5	54.3	64.3	68.8	71.5	69.9	67.2	62.4	51.5	43.5
	10%	DB	38.1	39.0	52.1	63.5	75.2	81.2	83.8	81.3	77.0	67.0	54.1	42.5
		MCWB	35.9	35.3	45.8	51.8	61.8	66.9	69.7	68.6	66.1	59.4	48.4	39.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	53.2	63.1	63.8	72.2	75.4	78.3	77.3	75.9	70.8	59.7	56.2
		MCDB	54.4	56.1	74.6	73.6	82.5	86.0	88.1	84.3	83.6	78.7	64.8	58.7
	2%	WB	46.2	46.0	57.1	59.7	69.6	73.3	76.1	74.5	73.0	67.6	55.4	48.7
		MCDB	47.6	50.2	62.9	68.2	78.2	83.8	85.0	80.9	79.5	75.0	61.3	50.6
	5%	WB	40.0	39.8	52.1	56.2	67.5	71.5	74.1	72.6	70.9	63.9	51.8	43.8
		MCDB	42.8	43.0	57.5	66.5	76.9	81.4	82.4	79.5	76.6	68.2	56.9	46.0
	10%	WB	36.3	35.5	45.7	53.0	64.2	69.6	72.4	71.1	68.5	60.1	48.3	39.2
		MCDB	38.3	38.6	51.1	62.0	72.4	77.4	80.2	78.3	74.1	65.9	53.4	41.9
Mean Daily Temperature Range	5% DB	MDBR	12.8	14.4	16.9	20.5	21.0	20.3	20.7	20.2	20.8	18.1	14.5	10.9
		MCDBR	16.1	17.7	23.9	28.8	24.1	22.4	22.9	22.6	23.5	25.0	21.1	16.4
		MCWBR	14.0	14.6	15.8	15.2	11.2	9.7	9.9	9.8	10.6	13.6	14.1	13.8
	5% WB	MCDBR	15.3	16.8	22.6	24.2	21.2	18.9	19.3	18.2	18.6	21.1	18.9	15.7
		MCWBR	14.0	14.8	16.6	15.9	11.3	10.1	9.7	9.4	10.5	14.3	14.9	14.5
Clear-Sky Solar Irradiance	taub	0.294	0.327	0.338	0.379	0.405	0.430	0.417	0.411	0.378	0.344	0.314	0.292	
	taud	2.365	2.259	2.368	2.275	2.257	2.219	2.264	2.273	2.357	2.449	2.477	2.425	
	Ebn at Noon	266	273	286	281	276	268	270	268	269	266	259	258	
	Edh at Noon	25	33	34	41	43	44	42	40	34	28	23	22	
All-Sky Solar Radiation	RadAvg	460	741	1135	1451	1698	1914	1934	1655	1329	845	519	364	
	RadStd	42	92	118	128	153	115	94	73	125	82	67	33	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page